II order systems

Consider the second order mechanical system represented by spring, mass, and dashpot. We had derived the differential equation governing the dynamics of the system as

$m\ddot{x} + b\dot{x} + kx = F(t)$……………………………(1)

This equation is a second order, ordinary, non-homogeneous, linear differential equation with constant coefficients. Since it is non-homogeneous equation, the solution will have two parts- a homogeneous solution (from the corresponding homogeneous part) and a particular solution (depending upon the nature of the particular type of input F(t)). The homogeneous solution is known as the natural response and the particular solution is known as the forced response.

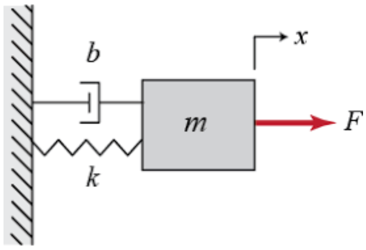


Fig 5. Second order mechanical system

Consider the corresponding homogeneous equation

$m\ddot{x} + b\dot{x} + kx = 0$ …………………………..(2)

Since the equation is linear, any exponential function will be a possible solution. Assuming a solution

$x(t) = Ae^{pt}$ ……………………………………….(3) where A is an arbitrary constant which can be evaluated from the initial conditions and p is a constant but not arbitrary. Substituting (3) in (2), we get

$mp^2Ae^{pt} + bpAe^{pt} + kAe^{pt} = 0$

$Ae^{pt}(mp^2 + bp + k) = 0,$

A or e^{pt} can not be equal to zero as it gives $x(t) = 0$ for all values of time which is a trivial solution. Hence for (3) to be a solution of (2), the condition is

$(mp^2 + bp + k) = 0$ ……………………………(4)

This is known as the characteristic equation, because the solution of it is going to give some particular characteristics of the system (which is p in this case).

$p = \frac{-b \pm \sqrt{b^2 - 4km}}{2m}$ ……………………………………(5)

We have two roots for p. Hence $x_1(t) = A_1e^{p_1t}$ and $x_2(t) = A_2e^{p_2t}$ are possible solutions of (2). Since the equation (2) is of second order, the solution should have two arbitrary constants. Because of the fact that the equation is linear, we can apply the principle of superpostion and any linear combination of the two solutions will be the possible solution.

Hence the homogeneous solution is

$x(t) = A_1e^{p_1t} + A_2e^{p_2t}$ …………………………(6) if the roots p_1 and p_2 are distinct (real or complex)

$x(t) = (A_1 + A_2t)e^{pt}$ ………………………………(7) if the roots are real and equal. (Refer your S_1 Calculus)

Depending on the value of the discriminant $\sqrt{b^2 - 4km}$ of (5), we have three cases

Case(i) $b^2 > 4km$

In this case, the roots are real and distinct. The effect of damping force (b) is more predominant than the spring force (k) and inertia force(m). This system is called an overdamped system.

Case(ii) $b^2 < 4km$

In this case, the roots are complex conjugates. The damping effect is less predominant in this system, which is called an underdamped system.

Case(iii) $b^2 = 4km$

Here, the roots are real and equal. This is called a critically damped system. Of the three cases, this is the only equality expression. We can find out the critical damping coefficient of the system as

$b_c = \sqrt{4km} = 2m\sqrt{\frac{k}{m}} = 2m\omega_n$ ....(8) where $\omega_n = \sqrt{\frac{k}{m}}$ is the undamped natural frequency of the system .

So, for every given system, we can calculate the critical damping coefficient associated with that system because it involves only the system properties $\boldsymbol{m}$ and $\boldsymbol{\omega_n}$ (or $\boldsymbol{m}$ and $\boldsymbol{k}$). But the actual damping of the system can be different.

For example; consider a mass 5 kg which is suspended with a spring of stiffness 1000 N/m. The critical damping coefficient of the system can be calculated as

$b_c = \sqrt{4km} = \sqrt{4 \times 1000 \times 5} = 141.4\ N/ms^{-1}$ ($\omega_n = \sqrt{\frac{1000}{5}} = 14.14\ rad/s, b_c = 2\text{x}5\text{x}14.14 = 141.4\ N/ms^{-1}$)

Suppose, it is allowed to vibrate in three media viz. air, water, and oil. Then, the actual damping present in the system will be different in these three cases (depending on their viscosity) which may not be equal to the critical damping we have calculated above. The actual damping can be more, equal, or less than the critical damping. Hence, we can always represent the actual damping present in a given system as a factor of the critical damping coefficient of that system (which can be evaluated from the system properties)

$b = \zeta\, b_c$ where ζ(zeta) is known as the damping ratio.

$$\zeta = \frac{actual\ damping\ coefficient\ of\ the\ system}{critical\ damping\ coeffient\ of\ the\ system}$$

Therefore, the actual damping coefficient of the system

$b = \zeta\, b_c = 2m\zeta\omega_n$...................................(9)

If ζ=1, the system is critically damped.

If ζ>1, the system is overdamped.

If ζ<1, the system is underdamped.

Now, (5) can be written as

$$p_{1,2} = \frac{-b}{2m} \pm \frac{\sqrt{b^2-4km}}{2m} = \frac{-b}{2m} \pm \sqrt{\left(\frac{b}{2m}\right)^2 - \left(\frac{k}{m}\right)} = -\zeta\omega_n \pm \sqrt{(\zeta\omega_n)^2 - (\omega_n)^2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2-1} \quad ...(10)$$

If $\zeta > 1$, the roots p_1 and p_2 are real and distinct. Then the homogenous solution (6) can be written as

$$\boldsymbol{x}(t) = A_1 e^{\left(-\zeta\omega_n + \omega_n\sqrt{\zeta^2-1}\right)t} + A_2 e^{\left(-\zeta\omega_n - \omega_n\sqrt{\zeta^2-1}\right)t}$$

$$x(t) = e^{-\zeta\omega_n t}(A_1 e^{(\omega_n\sqrt{\zeta^2-1})t} + A_2 e^{(-\omega_n\sqrt{\zeta^2-1})t} \quad ....(11)$$

If $\zeta = 1$, the roots p_1 and p_2 are real and equal ($p_{1,2} = -\omega_n$). Then the homogenous solution **(7)** can be written as

$$x(t) = (A_1 + A_2 t)e^{-\omega_n t} \quad ....(12)$$

If $\zeta < 1$, the roots p_1 and p_2 are complex conjugates.

$p_{1,2} = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2} = -\zeta\omega_n \pm j\omega_d$ where damped frequency $\omega_d = \omega_n\sqrt{1-\zeta^2}$. Then the homogenous solution (6) can be written as

$x(t) = A_1 e^{(-\zeta\omega_n + j\omega_d)t} + A_2 e^{(-\zeta\omega_n - j\omega_d)t} = e^{-\zeta\omega_n t}(A_1 e^{j\omega_d t} + A_2 e^{-j\omega_d t})$............(13a)

$= e^{-\zeta\omega_n t}\{A_1(\cos\omega_d t + j\sin\omega_d t) + A_2(\cos\omega_d t - j\sin\omega_d t)\}$

$= e^{-\zeta\omega_n t}\{(A_1 + A_2)\cos\omega_d t + j(A_1 - A_2)\sin\omega_d t\}$

$= e^{-\zeta\omega_n t}(C_1\cos\omega_d t + C_2\sin\omega_d t)$............(13b) C_1 and C_2 are the two new arbitrary constants

$= e^{-\zeta\omega_n t}(X_0\sin\theta\cos\omega_d t + X_0\cos\theta\sin\omega_d t)$ where $C_1 = X_0\sin\theta$ and $C_2 = X_0\cos\theta$ so that

$X_0 = \sqrt{C_1^2 + C_2^2}$ and $\theta = \tan^{-1}\frac{C_1}{C_2}$

i.e, $x(t) = X_0 e^{-\zeta\omega_n t}\sin(\omega_d t + \theta)$.................(14) where X_0 and θ are the new arbitrary constants.

The response of an underdamped system can be represented by any one of the three equations (13a), (13b), or (14).

Now we have obtained the natural response (homogeneous solution) of a second order system

$x(t) = e^{-\zeta\omega_n t}(A_1 e^{(\omega_n\sqrt{\zeta^2-1})t} + A_2 e^{(-\omega_n\sqrt{\zeta^2-1})t}$ if the system is overdamped ($\zeta > 1$)

$x(t) = (A_1 + A_2 t)e^{-\omega_n t}$ if the system is critically damped ($\zeta = 1$)

$x(t) = X_0 e^{-\zeta\omega_n t}\sin(\omega_d t + \theta)$ if the system is underdamped $\zeta < 1$)

The values of the arbitrary constants have to be determined from the initial conditions.

If we look at the first expression above for the overdamped system, irrespective of the values of the arbitrary constants as $t \to \infty$,

$e^{-\zeta\omega_n t} \to 0$, $e^{(\omega_n\sqrt{\zeta^2-1})t} \to \infty$, $e^{(-\omega_n\sqrt{\zeta^2-1})t} \to 0$

hence $x(t) \to 0$

In second expression of critical damped system as $t \to \infty$,

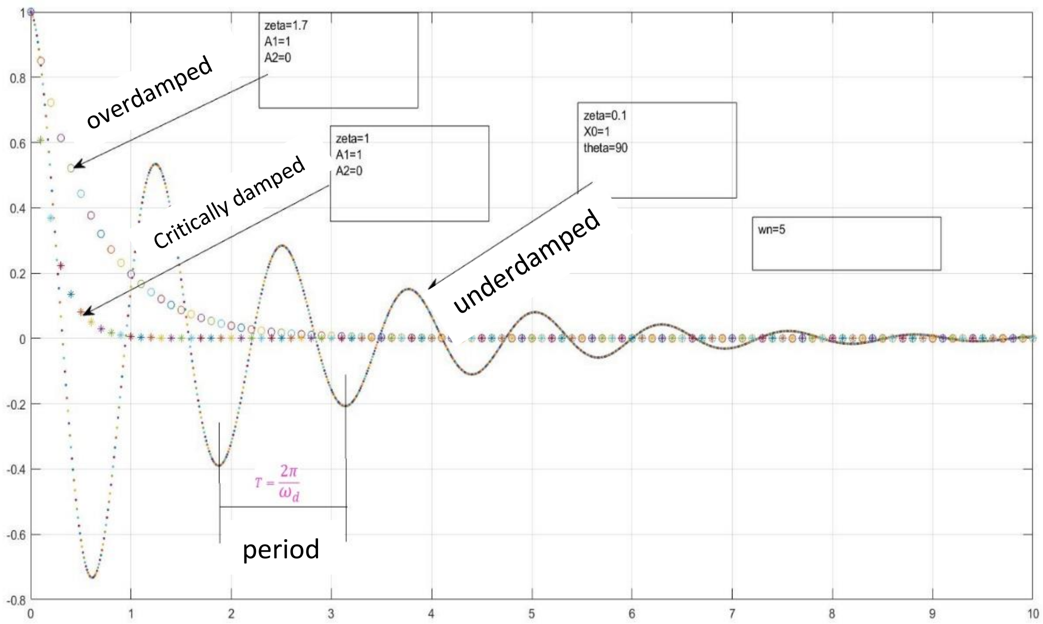
$(A_1 + A_2 t) \to \infty$, $e^{-\omega_n t} \to 0$, hence $x(t) \to 0$

In both the above cases, the system will not oscillate. Once the vibration is initiated by giving some initial conditions, the system comes to equilibrium without oscillation. The time taken by the system to come to equilibrium is smaller for critically damped system than an overdamped.

For the under damped system, the response is a sine curve with amplitude $X_0 e^{-\zeta\omega_n t}$ and circular frequency ω_d. The period of oscillation $T = \frac{2\pi}{\omega_d}$. The amplitude of oscillation $X_0 e^{-\zeta\omega_n t}$ decays exponentially to zero as $t \to \infty$.

Only an underdamped sytem will vibrate/oscillate when it is disturbed from equilibrium.

The homogeneous solutions for the three different cases are plotted in the figure below. The arbitrary constants are assumed as $A_1 = 1$, $A_2 = 0$, $X_0 = 1$, and $\theta = 90^0$ and $\omega_n = 5$



We can see that irrespctive of the values of damping factor, the natural response vanishes after some time.

The particular solution is obtained by considering a particular input $F(t)$. The response that we get is known as the forced response which will exist as long as $F(t)$ exists. We will consider the forced responses for step and impulse inputs later. The complete solution (total response) will be the sum of natural response(homogeneous solution) and forced response (particular solution)

#Exercise 1

Find the values of the arbitrary constants in the expressions for the homogeneous responses of overdamped ($\zeta = 1.7$), critically damped ($\zeta = 1$), and underdamped ($\zeta = 0.7$) with undamped natural frequency ($\omega_n = 5\, rad/s$) for the following set of initial conditions

(i) $x(0) = 1,\ \dot{x}(0) = 0$ (ii) $x(0) = 0,\ \dot{x}(0) = 1$ (iii) $x(0) = 1,\ \dot{x}(0) = 1$

Plot the responses using Matlab/Scilab/Excel

(Hint: For substituting the velocity initial condition, you need to differentiate the expression for x(t) with respect to time.)

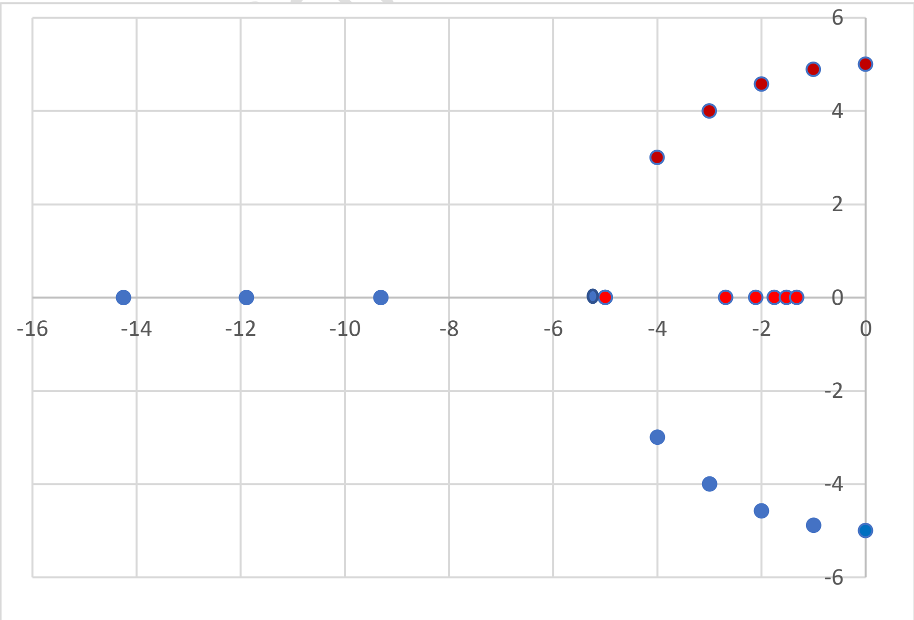
Let us again consider the roots of the characteristic equation given in (10)

$p_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$

Let us vary the value of ζ keeping ω_n as constant $(= 5\, rad/s)$

ζ	$-\zeta\omega_n$	$\omega_n\sqrt{\zeta^2-1}$	$p_1 = -\zeta\omega_n + \omega_n\sqrt{\zeta^2-1}$	$p_2 = -\zeta\omega_n - \omega_n\sqrt{\zeta^2-1}$
0	0	j5.00	+j5.0	-j5.0
0.2	-1.0	j4.89	-1.0+j4.89	-1.0-j4.89
0.4	-2.0	j4.58	-2.0+j4.58	-2.0-j4.58
0.6	-3.0	j4.00	-3.0+j4.00	-3.0-j4.00
0.8	-4.0	j3.00	-4.0+j3.00	-4.0-j3.00
1.0	-5.0	0	-5.0	-5.0
1.2	-6.0	3.31	-2.69	-9.31
1.4	-7.0	4.89	-2.11	-11.89
1.6	-8.0	6.25	-1.75	-14.25

Look at the values of p_1 and p_2 in the above table. It gives the values of the roots of the characteristic equation when ζ is varied from 0. The values of the roots are plotted in complex plane in figure below. The horizontal axis is the real axis and vertical one is the imaginary axis. When $\zeta = 0$ (undamped), the roots are $\pm j5$ which are on the imaginary axis. As we increase the value of $\zeta\ to\ 0.2$ (underdamped), the roots are complex conjugates with negative real part and shift to the left side of the imaginary axis. The values of root p_1 are marked in red dots and p_2 are shown in blue dots. When ζ =1 (critically damped), both the roots are equal at -5. When ζ increases above 1(overdamped), both the roots are negative real numbers. One moves to a large negative value and the other approaching to zero. Please note that the value of p_1 will never be equal to zero however high the value of ζ may be, because the negative part $-\zeta\omega_n$ will be always greater than the positive part $\omega_n\sqrt{\zeta^2-1}$. If we continuously change the parameter ζ and plot the roots of the charcteristic equation, we will get the locus of the roots of the characteristic equation which is known as the ROOT LOCUS. We will discuss the root locus technique in detail later.



If you have understood this far, let us approach this in another way.

The governing equation of a second order dynamic mechanical system is

$m\ddot{x} + b\dot{x} + kx = F(t)$.

Taking Laplace transform on both sides

$m\{s^2X(s) - sx(0) - \dot{x}(0)\} + b\{sX(s) - x(0)\} + kX(s) = F(s)$

$(ms^2 + bs + k)X(s) - \{(ms + b)x(0) + m\dot{x}(0)\} = F(s)$ ....(15)

$(ms^2 + bs + k)X(s) = \{(ms + b)x(0) + m\dot{x}(0)\} + F(s)$

$X(s) = \frac{\{(ms+b)x(0)+m\dot{x}(0)\}+F(s)}{ms^2+bs+k}$

$= \frac{\{(ms+b)x(0)+m\dot{x}(0)\}}{ms^2+bs+k} + \frac{F(s)}{ms^2+bs+k}$ ....(16)

The first term on RHS will give the natural response (homogeneous solution) and the second term will give the forced response (particular solution)

If we know the initial conditions and $F(s)$ {which is the Laplace Transform of the input $F(t)$ }, we can find out the response $x(t)$ by taking the inverse Laplace Transform of (16)

Assume that no input acts on the system $[F(s) = 0]$ and initial conditions $x(0) = 1,\ \ and\ \ \dot{x}(0) = 0$

$X(s) = \frac{\{(ms + b)x(0) + m\dot{x}(0)\}}{ms^2 + bs + k} \dots\dots\dots\dots\dots\dots\dots\dots\dots\dots\dots .(17)$

$= \frac{\{(ms+b).1+m.0\}}{ms^2+bs+k} = \frac{s+\frac{b}{m}}{s^2+\frac{b}{m}s+\frac{k}{m}} = \frac{s+2\zeta\omega_n}{s^2+2\zeta\omega_n s+\omega_n{}^2}$ (Refer Eq.(8) and (9))

$x(t)$ can be calculated by taking the invers transform. Let us write $X(s)$ as partial fraction

$X(s) = \frac{s+2\zeta\omega_n}{(s+\zeta\omega_n)^2-(\zeta\omega_n)^2+\omega_n{}^2} = \frac{s+2\zeta\omega_n}{[s+\zeta\omega_n]^2-[\omega_n\sqrt{\zeta^2-1}]^2}$ .... (18)

Let us assume that the system is critically damped ($\zeta = 1$), The equation (18) reduces to

$X(s) = \frac{s + 2\omega_n}{(s + \omega_n)^2} = \frac{A}{s + \omega_n} + \frac{B}{(s + \omega_n)^2}$ $\quad (Partial\ Fraction)$

$s + 2\omega_n = A(s + \omega_n) + B$..(19a)

By letting $s = -\omega_n$, we get $B = \omega_n$

Equating constants on both sides of (19a), we can write $2\omega_n = A\omega_n + B$

We get $A = 1$

$X(s) = \frac{1}{s + \omega_n} + \frac{\omega_n}{(s + \omega_n)^2}$

Taking inverse Laplace Transform

$x(t) = e^{-\omega_n t} + \omega_n(e^{-\omega_n t})\,t = e^{-\omega_n t}(1 + \omega_n t) \dots\dots\dots\dots\dots\dots\dots .(19b)$

This is exactly same as (12) with $A_1 = 1, A_2 = \omega_n$

(You will be getting the same expression as (19b) when you do #Exercise 1 for the critically damped system with the same initial conditions. Verify!!!!!!)

Let us consider an overdamped system ($\zeta > 1$). The equation (18) is written as (for simplification)

$X(s) = \frac{s+2a}{(s+a)^2-b^2}$ where $a = \zeta\omega_n$ and $b = \omega_n\sqrt{\zeta^2-1}$

$$X(s) = \frac{s+2a}{\{(s+a)+b\}\{(s+a)-b\}} = \frac{A}{s+a+b} + \frac{B}{s+a-b}$$

$s + 2a = A(s+a-b) + B(s+a+b)$. Let us put $s = b - a$ and $s = -a - b$ to evaluate A and B

$$b - a + 2a = A(b-a+a-b) + B(b-a+a+b) \quad \gg \quad B = \frac{a+b}{2b}$$

$$-a - b + 2a = A(-a-b+a-b) + B(-a-b+a+b) \quad \gg \quad A = -\frac{a-b}{2b}$$

$$X(s) = \frac{-\frac{a-b}{2b}}{s+a+b} + \frac{\frac{a+b}{2b}}{s+a-b} = \frac{1}{2b}\left\{\frac{a+b}{s+(a-b)} - \frac{a-b}{s+(a+b)}\right\}$$

Taking inverse Laplace Transform

$$x(t) = \frac{1}{2b}\left\{(a+b)\, e^{-(a-b)t} - (a-b)\, e^{-(a+b)t}\right\}$$

$$x(t) = \frac{1}{2\omega_n\sqrt{\zeta^2-1}}\left\{(\zeta\omega_n + \omega_n\sqrt{\zeta^2-1})\, e^{-\left(\zeta\omega_n - \omega_n\sqrt{\zeta^2-1}\right)t} - (\zeta\omega_n - \omega_n\sqrt{\zeta^2-1})\, e^{-\left(\zeta\omega_n + \omega_n\sqrt{\zeta^2-1}\right)t}\right\}$$

$$x(t) = \frac{e^{-(\zeta\omega_n)t}}{2\omega_n\sqrt{\zeta^2-1}}\left\{(\zeta\omega_n + \omega_n\sqrt{\zeta^2-1})\, e^{\left(\omega_n\sqrt{\zeta^2-1}\right)t} - (\zeta\omega_n - \omega_n\sqrt{\zeta^2-1})\, e^{-\left(\omega_n\sqrt{\zeta^2-1}\right)t}\right\}$$

$$x(t) = e^{-\zeta\omega_n t}\left\{\frac{\zeta+\sqrt{\zeta^2-1)}}{2\sqrt{\zeta^2-1}}\, e^{\left(\omega_n\sqrt{\zeta^2-1}\right)t} - \frac{\zeta-\sqrt{\zeta^2-1)}}{2\sqrt{\zeta^2-1}}\, e^{-\left(\omega_n\sqrt{\zeta^2-1}\right)t}\right\} \text{.....................(20)}$$

Compare this with equation (11) and your #Exercise 1 problem of overdamped system response with same initial conditions. Verify the value of A_1 and A_2 !!!!!

Consider now an underdamped system with $\zeta < 1$, the equation (18) is written as

$$X(s) = \frac{s+2\zeta\omega_n}{[s+\zeta\omega_n]^2-[\omega_n\sqrt{\zeta^2-1}]^2} = \frac{s+2\zeta\omega_n}{[s+\zeta\omega_n]^2-[j\omega_n\sqrt{1-\zeta^2}]^2} = \frac{(s+\zeta\omega_n)+\zeta\omega_n}{[s+\zeta\omega_n]^2+[\omega_d]^2}$$

Recall that $\omega_d = \omega_n\sqrt{1-\zeta^2}$ and $j^2=(-1)$

$$X(s) = \frac{(s+\zeta\omega_n)}{[s+\zeta\omega_n]^2+[\omega_d]^2} + \frac{\zeta\omega_n}{[s+\zeta\omega_n]^2+[\omega_d]^2} = \frac{(s+\zeta\omega_n)}{[s+\zeta\omega_n]^2+[\omega_d]^2} + \left(\frac{\zeta\omega_n}{\omega_d}\right)\frac{\omega_d}{[s+\zeta\omega_n]^2+[\omega_d]^2}$$

Taking inverse Laplace Transform (revisit the laplace transforms of sin wt and cos wt)

$$x(t) = e^{-\zeta\omega_n t}\cos\omega_d t + \left(\frac{\zeta\omega_n}{\omega_d}\right)e^{-\zeta\omega_n t}\sin\omega_d t = e^{-\zeta\omega_n t}\left(\cos\omega_d t + \left(\frac{\zeta}{\sqrt{1-\zeta^2}}\right)\sin\omega_d t\right)$$

$$= \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\left(\sqrt{1-\zeta^2}\cos\omega_d t + \zeta\sin\omega_d t\right) = \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}(X_0\sin\theta\cos\omega_d t + X_0\cos\theta\sin\omega_d t)$$

where $X_0\sin\theta = \sqrt{1-\zeta^2}$ *and* $X_0\cos\theta = \zeta$ so that $X_0{}^2 = (1-\zeta^2) + \zeta^2 = 1$ and $\tan\theta = \frac{\sqrt{1-\zeta^2}}{\zeta}$

$$x(t) = \frac{1}{\sqrt{1-\zeta^2}}\, e^{-\zeta\omega_n t}\, \sin(\omega_d t + \theta) \text{ ....(21)}$$

Compare this with equation (14) and your #Exercise 1 problem of underdamped system response with same initial conditions. Verify the value of X_0 and θ !!!!!

#Exercise 2.
Starting from Equation (17), evaluate the natural responses of over, under, and critically damped second order systems for the following initial conditions. Verify your answer with #Ex. 1. (i) $x(0) = 0,\ \dot{x}(0) = 1$ (ii) $x(0) = 1,\ \dot{x}(0) = 1$

If we set all initial conditions equal to zero, (16) reduces to

$X(s) = \frac{F(s)}{ms^2+bs+k}$.

The transfer function of the system is given by (Recall the definition of Transfer Function)

$$G(s) = \frac{X(s)}{Y(s)} = \frac{1}{ms^2+bs+k} = \frac{1}{m(s^2+\frac{b}{m}s+\frac{k}{m})} = \left(\frac{1}{k}\right)\left(\frac{k}{m}\right)\frac{1}{(s^2+\frac{b}{m}s+\frac{k}{m})} = \left(\frac{1}{k}\right)\frac{\frac{k}{m}}{(s^2+\frac{b}{m}s+\frac{k}{m})}$$

$$G(s) = \left(\frac{1}{k}\right)\frac{{\omega_n}^2}{s^2+2\zeta\omega_n s+{\omega_n}^2}$$

This is now of the form $G(s) = K\frac{p(s)}{q(s)}$.

The standard form of a second order system is considered as $\frac{{\omega_n}^2}{s^2+2\zeta\omega_n s+{\omega_n}^2}$

The characteristic equation is $q(s) = 0$, and the roots of the characteristic equation are called poles of the system

$s^2 + 2\zeta\omega_n s + {\omega_n}^2 = 0$ ....(22)

$s = \frac{-2\zeta\omega_n \pm \sqrt{(2\zeta\omega_n)^2 - 4{\omega_n}^2}}{2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2-1}$ ....(23) [Same as equation (10)]

Example 1

The transfer function of a second order system is $G(s) = \frac{25}{s^2+5s+25}$ Find the natural frequency and damping ratio of the system.

Comparing the given transfer function with the standard form $\frac{{\omega_n}^2}{s^2+2\zeta\omega_n s+{\omega_n}^2}$

${\omega_n}^2 = 25$, and $2\zeta\omega_n = 5$

Hence $\omega_n = 5$ rad/s, and $\zeta = \frac{5}{2\omega_n} = 0.5$

#Exercise 3

Given below are the natural frequency and damping ratio of a second order system respectively. What is system transfer function? Where is the location of the poles?

i) 3 rad/s and 0.25 ii) 5 rad/s and 1 iii) 2 rad/s and 1.25 iv) 4 rad/s and 0

#Exercise 4

Find the natural frequency and damping ratio of the second oerder systems given below

i) $G(s) = \frac{24}{3s^2+8s+24}$ ii) $G(s) = \frac{15}{(s+3)(2s+5)}$

#Exercise 5

The poles of three second order systems are given below. Obtain their transfer functions, damping ratios and natural frequencies.

i) -1, -5 ii) -3,-3 iii) -2±j5 iv) ±j5

Having learned the natural responses of a second order system both by solving the differential equation in the conventional way and with Laplace transform, let us look at the forced responses using the Laplace Transform technique

Impulse Response of Second Order Systems

Let us consider a second order system represented by the transfer function $G(s) = \frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$

$$G(s) = \frac{X(s)}{F(s)} = \frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$$

$$X(s) = F(s)\frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$$

For a unit impulse $F(t) = \delta(t)$, $F(s) = 1$

$$X(s) = \frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2} = \frac{\omega_n^2}{(s+\zeta\omega_n)^2-(\zeta\omega_n)^2+\omega_n^2} = \frac{\omega_n^2}{[s+\zeta\omega_n]^2-[\omega_n\sqrt{\zeta^2-1}]^2} \quad \text{.............................(24)}$$

Critically damped system ($\zeta = 1$)

The equation (24) reduces as follows.

$$X(s) = \frac{\omega_n^2}{(s+\omega_n)^2} = \frac{A}{s+\omega_n} + \frac{B}{(s+\omega_n)^2} \qquad (Partial\ Fraction)$$

$$\omega_n^2 = A(s+\omega_n) + B \quad \text{...(24a)}$$

By letting $s = -\omega_n$, we get $B = \omega_n^2$

Equating constants on both sides of (24a), we can write $\omega_n^2 = A\omega_n + B$

We get $A = 0$

$$X(s) = \frac{\omega_n^2}{(s+\omega_n)^2}$$

Taking inverse Laplace Transform

$$x(t) = \omega_n^2\, t\, e^{-\omega_n t} \quad \text{..(25)}$$

Compare this equation with the natural response obtained for a critically damped second order system with the initial conditions $x(0) = 0,\ \dot{x}(0) = 1$ (#Exercise 1 and 2)

Overdamped system ($\zeta > 1$)

The equation (24) is written as (for simplification)

$$X(s) = \frac{\omega_n^2}{(s+a)^2-b^2} \qquad \text{where } a = \zeta\omega_n \text{ and } b = \omega_n\sqrt{\zeta^2-1}$$

$$X(s) = \frac{\omega_n^2}{\{(s+a)+b\}\{(s+a)-b\}} = \frac{A}{s+a+b} + \frac{B}{s+a-b}$$

$\omega_n^2 = A(s+a-b) + B(s+a+b)$. Let us put $s = b-a$ and $s = -a-b$ to evaluate A and B

$$\omega_n^2 = A(b-a+a-b) + B(b-a+a+b) \qquad \gg\ B = \frac{\omega_n^2}{2b}$$

$$\omega_n^2 = A(-a-b+a-b) + B(-a-b+a+b) \qquad \gg\ A = -\frac{\omega_n^2}{2b}$$

$$X(s) = \frac{-\frac{\omega_n^2}{2b}}{s+a+b} + \frac{\frac{\omega_n^2}{2b}}{s+a-b} = \frac{\omega_n^2}{2b}\left\{\frac{1}{s+(a-b)} - \frac{1}{s+(a+b)}\right\}$$

Taking inverse Laplace Transform

$$x(t) = \frac{\omega_n^2}{2b}\left\{e^{-(a-b)t} - e^{-(a+b)t}\right\}$$

$$x(t) = \frac{\omega_n^2}{2\omega_n\sqrt{\zeta^2-1}}\left\{e^{-\left(\zeta\omega_n-\omega_n\sqrt{\zeta^2-1}\right)t} - e^{-\left(\zeta\omega_n+\omega_n\sqrt{\zeta^2-1}\right)t}\right\}$$

$$x(t) = \frac{\omega_n e^{-(\zeta\omega_n)t}}{2\sqrt{\zeta^2-1}}\left\{e^{\left(\omega_n\sqrt{\zeta^2-1}\right)t} - e^{-\left(\omega_n\sqrt{\zeta^2-1}\right)t}\right\}$$

$$x(t) = e^{-\zeta\omega_n t}\left\{\frac{\omega_n}{2\sqrt{\zeta^2-1}}e^{\left(\omega_n\sqrt{\zeta^2-1}\right)t} - \frac{\omega_n}{2\sqrt{\zeta^2-1}}e^{-\left(\omega_n\sqrt{\zeta^2-1}\right)t}\right\} \quad \text{......................(26)}$$

Compare this equation with the natural response obtained for a overdamped second order system with the initial conditions $x(0) = 0,\ \dot{x}(0) = 1$ (#Exercise 1 and 2)

Underdamped system ($\zeta < 1$)

The equation (24) is written as

$$X(s) = \frac{{\omega_n}^2}{[s+\zeta\omega_n]^2-[\omega_n\sqrt{\zeta^2-1}]^2} = \frac{{\omega_n}^2}{[s+\zeta\omega_n]^2-[j\omega_n\sqrt{1-\zeta^2}]^2} = \frac{{\omega_n}^2}{[s+\zeta\omega_n]^2+[\omega_d]^2}$$

Recall that $\omega_d = \omega_n\sqrt{1-\zeta^2}$ and $j^2=(-1)$

$$X(s) = \left(\frac{{\omega_n}^2}{\omega_d}\right)\frac{\omega_d}{[s+\zeta\omega_n]^2+[\omega_d]^2}$$

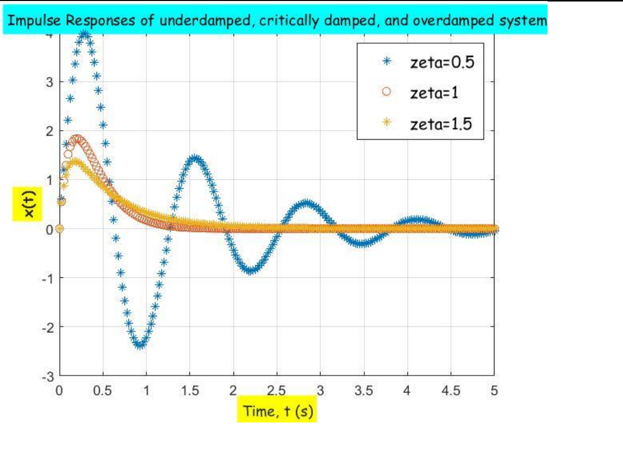
Taking inverse Laplace Transform (revisit the laplace transforms of sin wt)

$$x(t) = \left(\frac{{\omega_n}^2}{\omega_d}\right)e^{-\zeta\omega_n t}\sin\omega_d t$$

$$x(t) = \frac{\omega_n}{\sqrt{1-\zeta^2}}\ e^{-\zeta\omega_n t}\ \sin\omega_d t \text{ ....(27)}$$

Compare this equation with the natural response obtained for an underdamped second order system with the initial conditions $x(0) = 0,\ \dot{x}(0) = 1$ (#Exercise 1 and 2)

The impulse responses of a second order system with ω_n=5 rad/s and ζ values of 0.5, 1, and 1.5 are shown in Figure. The behavior of the system when an impulse is given as the input (forced response/particular solution) is similar to its natural response (homogeneous solution) with the initial conditions $x(0) = 0$, $\dot{x}(0) = 1$. This is understandable because the impulse acts only for a very short duration initially and thereafter the system behaves naturally as there is no input/forcing term.



#Exercise 6

Given below are the natural frequency and damping ratio of a second order system respectively. Plot the impulse responses using Matlab/Scilab.

ii) 3 rad/s and 0.25 ii) 5 rad/s and 1 iii) 2 rad/s and 1.25 iv) 4 rad/s and 0

#Exercise 7

The transfer functions of second order system are given below. Plot the impulse responses using Matlab/Scilab

i) $G(s) = \frac{24}{3s^2+8s+24}$ ii) $G(s) = \frac{15}{(s+3)(2s+5)}$

#Exercise 8

The poles of three second order systems are given below. Plot the impulse responses using Matlab/Scilab

i) -1, -5 ii) -3,-3 iii) -2±j5 iv) ±j5

Step Response of Second Order Systems

Consider a second order sytem with transfer function $G(s) = \frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$

$G(s) = \frac{X(s)}{F(s)} = \frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$

$X(s) = F(s)\frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$

For a unit step input $F(t) = 1$, $F(s) = 1/s$

$X(s) = \left(\frac{1}{s}\right)\frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2} = \left(\frac{1}{s}\right)\frac{\omega_n^2}{(s+\zeta\omega_n)^2-(\zeta\omega_n)^2+\omega_n^2} = \left(\frac{1}{s}\right)\frac{\omega_n^2}{[s+\zeta\omega_n]^2-[\omega_n\sqrt{\zeta^2-1}]^2}$ ....(27a)

Critically damped system ($\zeta = 1$)

The equation (27a) becomes

$X(s) = \left(\frac{1}{s}\right)\frac{\omega_n^2}{(s+\omega_n)^2} = \frac{A}{s} + \frac{B}{s+\omega_n} + \frac{C}{(s+\omega_n)^2}$ ....(28)

$\omega_n^2 = A(s+\omega_n)^2 + Bs(s+\omega_n) + Cs$

Substituting $s = 0$, we wet $A = 1$

When $s = -\omega_n$ ⋙ $\boldsymbol{C = -\omega_n}$

Equating coefficients of s^2, $0 = A + B$ ⋙ $B = -1$

Therefore $X(s) = \frac{1}{s} - \frac{1}{s+\omega_n} - \frac{\omega_n}{(s+\omega_n)^2}$

Taking inverse Laplace Transform

$x(t) = 1 - e^{-\omega_n t} - \omega_n t e^{-\omega_n t}$

$x(t) = 1 - (1+\omega_n t)e^{-\omega_n t}$ ....(29)

The response curve starts from origin (when $t = 0$) and reaches 1 as t→ ∞

> #Exercise 9
>
> Plot equation (29) in Matlab/Scilab/Excel for different values of ω_n=1, 3, and 5

Overdamped system ($\zeta > 1$)

We have equation (27)

$X(s) = \left(\frac{1}{s}\right)\frac{\omega_n^2}{[s+\zeta\omega_n]^2-[\omega_n\sqrt{\zeta^2-1}]^2}$

$X(s) = \left(\frac{1}{s}\right)\frac{\omega_n^2}{(s+a)^2-b^2}$ where $a = \zeta\omega_n$ and $b = \omega_n\sqrt{\zeta^2-1}$

$X(s) = \frac{\omega_n^2}{s\{(s+a)+b\}\{(s+a)-b\}} = \frac{A}{s+a+b} + \frac{B}{s+a-b} + \frac{C}{s}$

$\omega_n^2 = A\,s(s+a-b) + B\,s(s+a+b) + C(s+a+b)(s+a-b)$.

Let us put $s = b-a$, $s = -a-b$, and $s = 0$ to evaluate B, A, and C respectively.

$\omega_n^2 = B(b-a)(b-a+a+b)$ ≫ $B = -\frac{\omega_n^2}{2b(a-b)}$

$\omega_n^2 = A(-a-b)(-a-b+a-b)$ ≫ $A = \frac{\omega_n^2}{2b(a+b)}$

$\omega_n^2 = C(a+b)(a-b)$ ≫ $C = \frac{\boldsymbol{\omega_n^2}}{a^2-b^2} = 1$ (note that $a^2 - b^2 = \boldsymbol{\omega_n^2}$)

$X(s) = \left(\frac{\omega_n^2}{2b(a+b)}\right)\frac{1}{s+a+b} - \left(\frac{\omega_n^2}{2b(a-b)}\right)\frac{1}{s+a-b} + \frac{1}{s}$

Taking inverse Laplace Transform

$$x(t) = \left\{\omega_n{}^2 \frac{e^{-(a+b)t}}{2b(a+b)} - \omega_n{}^2 \frac{e^{-(a-b)t}}{2b(a-b)} + 1\right\}$$

$$= 1 + \frac{\omega_n{}^2(a-b)e^{-(a+b)t} - \omega_n{}^2(a+b)e^{-(a-b)t}}{2b(a^2-b^2)}$$

$$= 1 + \frac{\omega_n{}^2(a-b)e^{-(a+b)t} - \omega_n{}^2(a+b)e^{-(a-b)t}}{2b\omega_n{}^2}$$

$$= 1 + \frac{(a-b)e^{-(a+b)t} - (a+b)e^{-(a-b)t}}{2b} = 1 + \left(\frac{e^{-at}}{2b}\right)\{(a-b)e^{-bt} - (a+b)e^{bt}\}$$

$$= 1 - \left(\frac{e^{-at}}{2b}\right)\{(a+b)e^{bt} - (a-b)e^{-bt}\}$$

$$x(t) = 1 - \frac{e^{-\zeta\omega_n t}}{2\omega_n\sqrt{\zeta^2-1}}\left[(\zeta\omega_n + \omega_n\sqrt{\zeta^2-1}\,)e^{\omega_n\sqrt{\zeta^2-1}\,t} - (\zeta\omega_n - \omega_n\sqrt{\zeta^2-1}\,)e^{-\omega_n\sqrt{\zeta^2-1}\,t}\right]$$

$$x(t) = 1 - e^{-\zeta\omega_n t}\left\{\left(\frac{\zeta+\sqrt{\zeta^2-1}}{2\sqrt{\zeta^2-1}}\right)e^{\omega_n\sqrt{\zeta^2-1}\,t} - \left(\frac{\zeta-\sqrt{\zeta^2-1}}{2\sqrt{\zeta^2-1}}\right)e^{-\omega_n\sqrt{\zeta^2-1}\,t}\right\} \text{.....................(30)}$$

#Exercise 10

Plot equation (30) in Matlab/Scilab/Excel with ζ =2 for different values of $\omega_n = 1, 3,$ and 5

Underdamped system ($\zeta < 1$)

Let us start from equation (27)

$$X(s) = \left(\frac{1}{s}\right)\frac{\omega_n{}^2}{[s+\zeta\omega_n]^2 - [\omega_n\sqrt{\zeta^2-1}]^2} = \left(\frac{1}{s}\right)\frac{\omega_n{}^2}{[s+\zeta\omega_n]^2 - [j\omega_n\sqrt{1-\zeta^2}]^2} = \left(\frac{1}{s}\right)\frac{\omega_n{}^2}{[s+\zeta\omega_n]^2 + [\omega_d]^2}$$

Recall that $\omega_d = \omega_n\sqrt{1-\zeta^2}$ and $j^2 = (-1)$

$X(s) = \left(\frac{1}{s}\right)\frac{\omega_n{}^2}{[s+\zeta\omega_n]^2+[\omega_d]^2} = \frac{A}{s} + \frac{Bs+C}{s^2+2\zeta\omega_n s+\omega_n{}^2}$ (since the numerator can not be factorised,it is kept as such)

$$\omega_n{}^2 = A(s^2 + 2\zeta\omega_n s + \omega_n{}^2) + (Bs + C)s$$

When $s = 0,\ A = 1$

Equating coefficients of s², $0 = A + B \quad ⋙ \quad B = -1$

Equating coefficients of s, $0 = A2\zeta\omega_n + C \quad ⋙ \quad C = -2\zeta\omega_n$

$$X(s) = \frac{1}{s} - \frac{s+2\zeta\omega_n}{s^2+2\zeta\omega_n s+\omega_n{}^2} = \frac{1}{s} - \frac{s+2\zeta\omega_n}{[s+\zeta\omega_n]^2+[\omega_d]^2}$$

$$= \frac{1}{s} - \frac{s+\zeta\omega_n}{[s+\zeta\omega_n]^2+[\omega_d]^2} - \frac{\zeta\omega_n}{[s+\zeta\omega_n]^2+[\omega_d]^2}$$

$$= \frac{1}{s} - \frac{s+\zeta\omega_n}{[s+\zeta\omega_n]^2+[\omega_d]^2} - \left(\frac{\zeta\omega_n}{\omega_d}\right)\frac{\omega_d}{[s+\zeta\omega_n]^2+[\omega_d]^2}$$

$$X(s) = \frac{1}{s} - \frac{s+\zeta\omega_n}{[s+\zeta\omega_n]^2+[\omega_d]^2} - \left(\frac{\zeta}{\sqrt{1-\zeta^2}}\right)\frac{\omega_d}{[s+\zeta\omega_n]^2+[\omega_d]^2}$$

Taking inverse Laplace Transform (revisit the laplace transforms of $\sin\omega t$ and $\cos\omega t$)

$$x(t) = 1 - e^{-\zeta\omega_n t}\cos\omega_d t - \left(\frac{\zeta}{\sqrt{1-\zeta^2}}\right)e^{-\zeta\omega_n t}\sin\omega_d t = 1 - e^{-\zeta\omega_n t}\left\{\cos\omega_d t + \left(\frac{\zeta}{\sqrt{1-\zeta^2}}\right)\sin\omega_d t\right\}$$

$$= 1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\left\{\sqrt{1-\zeta^2}\cos\omega_d t + \zeta\sin\omega_d t\right\}$$

$$=1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}(X_0\sin\theta\cos\omega_d t + X_0\cos\theta\sin\omega_d t)$$

where $X_0\sin\theta = \sqrt{1-\zeta^2}$ and $X_0\cos\theta = \zeta$

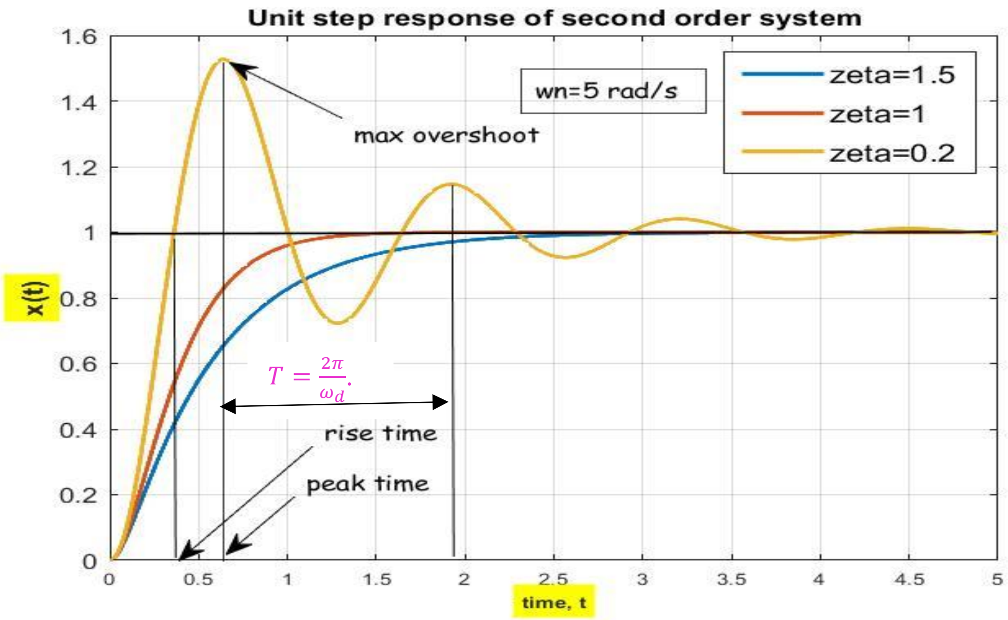
$X_0{}^2 = (1-\zeta^2) + \zeta^2 = 1$ and $\tan\theta = \frac{\sqrt{1-\zeta^2}}{\zeta}$

$$x(t) = 1 - \frac{1}{\sqrt{1-\zeta^2}}\,e^{-\zeta\omega_n t}\,\sin(\omega_d t + \theta) \text{ ....(31)}$$

#Exercise 11

Plot equation (31) in Matlab/Scilab/Excel with ζ =0.2 for different values of ω_n=1, 3, and 5. Note that the value of θ should be in radian.

The unit step responses of these systems are shown in figure below



Looking at the plots of equations (29), (30), and (31), we can understand that the responses of critically damped and overdamped systems reach the desired output without oscillation as t→ ∞, where as, an underdamped system response will have oscillations before attaining the final value. The frequency (circular frequency) of oscillations is ω_d rad/s with period $T = \frac{2\pi}{\omega_d}$. These oscillations die out after some time and the response reaches a steady state.

#Exercise 12

Using Matlab/Scilab, plot the step responses of a system with ζ =0.1 for different values of ω_n=1, 3, and 5. Repeat it with ω_n=5 for different values of ζ =0.1, 0.5, 0.8, 1, and 1.5. Examine the plots and draw your inferences.

We can see that the transient reponse and steady state response depends on the values of the poles of the second order transfer function $-\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$ given in Equaion(23). It is already explained that if the system is overdamped, the poles are negative real numbers given by

$-\zeta\omega_n - \omega_n\sqrt{\zeta^2 - 1}$ and $-\zeta\omega_n + \omega_n\sqrt{\zeta^2 - 1}$

If it is critically damped, we have a double pole on the real axis at $-\zeta\omega_n$. For underdamped system, the poles are complex conjugate with negative real part given by $-\zeta\omega_n \pm j\omega_n\sqrt{1 - \zeta^2}$

Let us consider a transfer function $G(s) = \frac{25}{s^2+2s+25}$. The poles are obtained from the characteristic equation $s^2 + 2s + 25 = 0$, the roots of which are $s = -1 \pm j4.9$. (We can also find the poles by knowing the values of $\zeta = 0.2$ *and* $\omega_n = 5$ from the transfer function sothat $-\zeta\omega_n = -1$ and $\omega_d = \omega_n\sqrt{1 - \zeta^2} = 4.9$)

We can plot these poles on the s-plane which shown in figure.

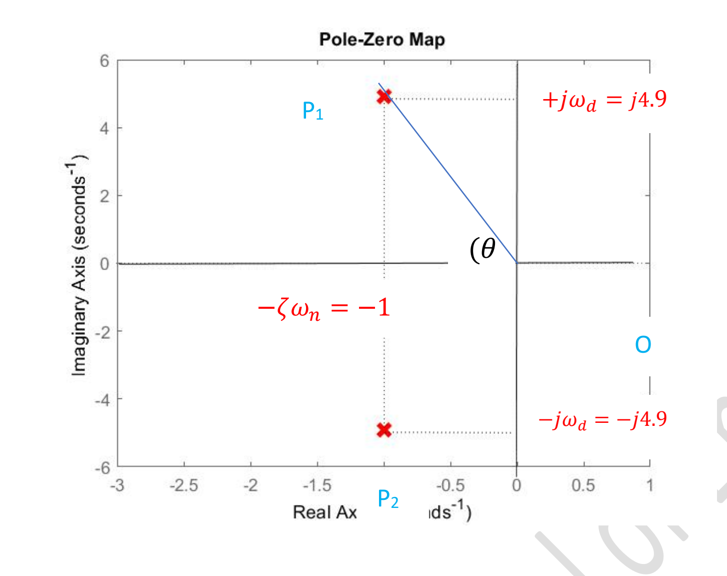
The distance $OP1 = \sqrt{(-\zeta\omega_n)^2 + (\omega_d)^2} = \sqrt{(\zeta\omega_n)^2 + \left(\omega_n\sqrt{1-\zeta^2}\right)^2} = \sqrt{\zeta^2{\omega_n}^2 + {\omega_n}^2(1-\zeta^2)} = \omega_n$

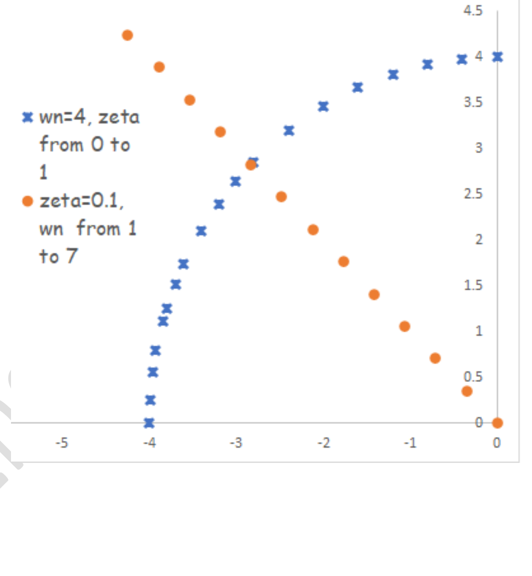
The angle θ can be found out using any of the following relations

$\tan\theta = \frac{\omega_n\sqrt{1-\zeta^2}}{\zeta\omega_n} = \frac{\sqrt{1-\zeta^2}}{\zeta}$ ..(32a)	$\cos\theta = \frac{\zeta\omega_n}{\omega_n} = \zeta$ ......(32b)	$\sin\theta = \frac{\omega_n\sqrt{1-\zeta^2}}{\omega_n} = \sqrt{1-\zeta^2}$..(32c)

Where ever be the pole location, the distance between the pole and the origin is always ω_n. If we fix ω_n and change the value of ζ, the pole location moves along the circular path as shown by the blue * in fig. 10 b.

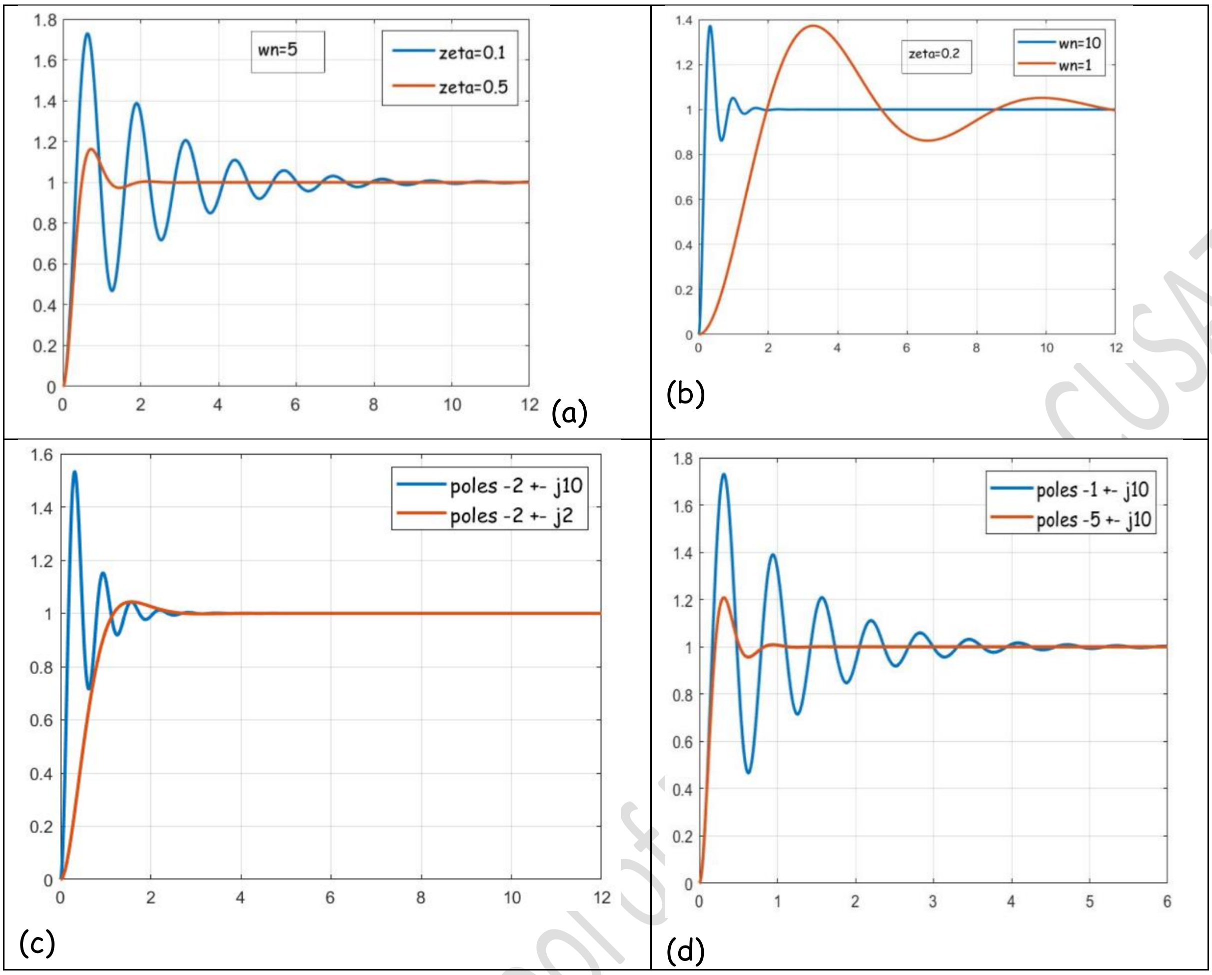
If we fix ζ and change the value of ω_n, the location of the pole moves along the line inclined at an angle $\theta = \cos^{-1}\zeta$ as shown by the brown in fig.10 b.





The pole-zero plot of $G(s) = \frac{25}{s^2+2s+25}$ b. movement of the pole location with ζ,ω_n

You might have observed the influence of ζ and ω_n (or rather the location of the poles) on the transient response while doing #Excersise 16. The figures below shows the influence of damping factor and natural frequency on the step response. What are your inferences?



Comparison of the step responses for different values of damping factor and frequency

Please remember that the type of transient respons is determined by the closed loop poles while the shape of the transient response is primarily determined by the closed loop zeroes.

We have already seen that depending on the location of the poles, the transient response can be undamped, underdamped, critically damped, or overdamped. Now see the influence of location of zero on the transient response.

Look at fig. 12. They represent the unit step responses of

i) $G(s) = \frac{25}{s^2+5s+25}$ (No zero)

ii) $G(s) = \frac{25s}{s^2+5s+25}$ (zero at z=0)

iii) $G(s) = \frac{25(s+1)}{s^2+5s+25}$ (zero at z= -1)

iv) $G(s) = \frac{25(s+4)}{s^2+5s+25}$ ((zero at z= -4)

Poles remain the same.

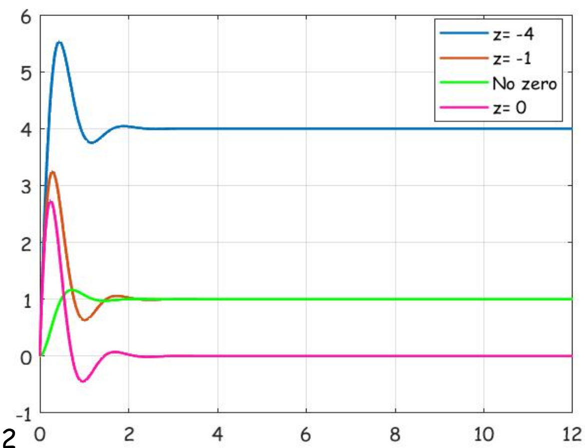


Fig.12

The transient behavior of an underdamped system can be specified by rise time, peak time, settling time, and maximum percentage overshoot.

Rise time (t_r): It is the time taken by the system to reach the desired output for the first time. When $t = t_r,\ x(t) = 1$. Therefore, Equation (31) can be written as

$$1 = 1 - \frac{1}{\sqrt{1-\zeta^2}}\ e^{-\zeta\omega_n t_r}\ \sin(\omega_d t_r + \theta)$$

$$\frac{1}{\sqrt{1-\zeta^2}}\ e^{-\zeta\omega_n t_r}\ \sin(\omega_d t_r + \theta) = 0$$

$$\sin(\omega_d t_r + \theta) = 0$$

$\omega_d t_r + \theta = 0, \pi, 2\pi \ldots\ldots$ (If we consider 0, we get negative time. Hence we take π)

$$t_r = \frac{\pi-\theta}{\omega_d} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots(33a)$$

where $\theta = \cos^{-1}\zeta$ and $\boldsymbol{\omega_d = \omega_n\sqrt{1-\zeta^2}}$

Peak Time (tp): It is the time taken by the system to reach the maximum (peak) value of the out put. When x(t) is maximum, $\frac{d}{dt}\ x(t) = 0$. Differentiating Equation (31) with repect to time

$$\frac{d}{dt}\ x(t) = -\frac{1}{\sqrt{1-\zeta^2}}\ e^{-\zeta\omega_n t_p}\ \cos(\omega_d t_p + \theta)\ \omega_d + \sin(\omega_d t_p + \theta)\left\{-\frac{1}{\sqrt{1-\zeta^2}}(-\zeta\omega_n)e^{-\zeta\omega_n t_p}\right\} = 0$$

$$\cos(\omega_d t_p + \theta)\ \omega_d = \sin(\omega_d t_p + \theta)\zeta\omega_n$$

$$\frac{\sin(\omega_d t_p+\theta)}{\cos(\omega_d t_p+\theta)} = \frac{\omega_d}{\zeta\omega_n} = \frac{\sqrt{1-\zeta^2}}{\zeta} = \tan\theta \quad \text{(from 32a)}$$

$$\tan(\omega_d t_p + \theta) = \tan\theta = \tan(\pi + \theta)$$

$$\omega_d t_p + \theta = \pi + \theta$$

$$t_p = \frac{\pi}{\omega_d} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots(33b)$$

Maximum overshoot: It is the maximum value reached by the system response above the desired output. When $t = t_p$, $x(t_p)$ is maximum

$$x(t_p) = 1 - \frac{1}{\sqrt{1-\zeta^2}}\left(e^{-\zeta\omega_n t_p}\right)\sin(\omega_d t_p + \theta) = 1 - \frac{1}{\sqrt{1-\zeta^2}}\left(e^{-\zeta\omega_n\frac{\pi}{\omega_d}}\right)\sin\left(\omega_d\frac{\pi}{\omega_d} + \theta\right)$$

$$= 1 - \frac{1}{\sqrt{1-\zeta^2}}\left(e^{-\zeta\omega_n\frac{\pi}{\omega_n\sqrt{1-\zeta^2}}}\right)\sin(\pi + \theta) = 1 + \frac{1}{\sqrt{1-\zeta^2}}\left(e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}}\right)\sin\theta$$

$$= 1 + \frac{1}{\sqrt{1-\zeta^2}}\left(e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}}\right)\sqrt{1-\zeta^2} = 1 + e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}} \quad \text{(from 32 c)}$$

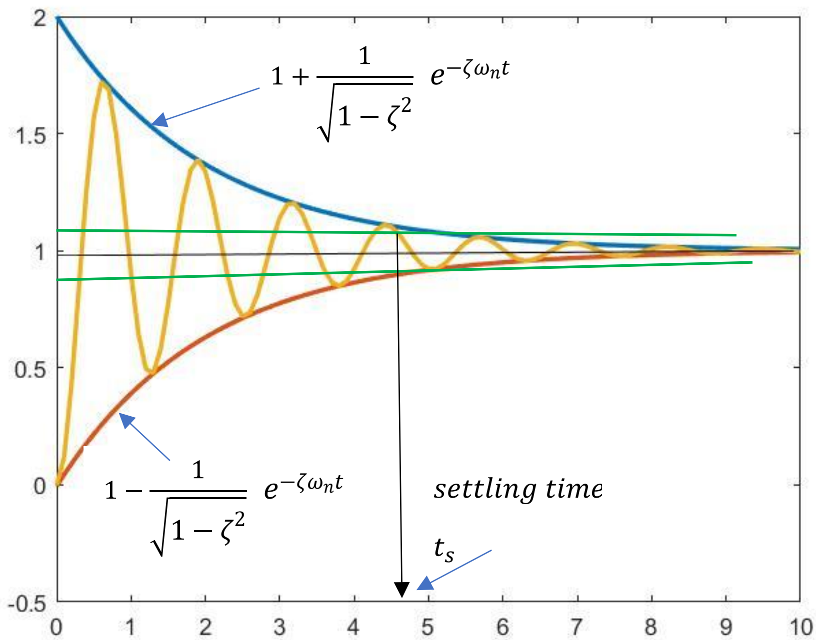
Maximum overshoot = $x(t_p) - desired\ output = \left(1 + e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}}\right) - 1 = e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}}$

Max.overshoot= $e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}}$ ……………………….(33c) (it is independent of ω_n)

Settling time (t_s): The amplitude of the underdamped response will be always between the envelopes $1 \pm \frac{1}{\sqrt{1-\zeta^2}}\ e^{-\zeta\omega_n t}$as shown in figure 11.Time constant of these envelopes is $\frac{1}{\zeta\omega_n}$. The speed of response depends on the value of this time constant. The settling time is defined as the time required for the system to settle within a certain percentage (usually $\pm 2\%$) of the desired output. This $\pm 2\%$ band is shown by the green line in fig.11. Settling time is the time taken by the system after which the output completely remains within the band. This happens approximately when $e^{-\zeta\omega_n t_s} < 0.02$

Or $t_s \cong \frac{4}{\zeta\omega_n}$ ……………………………….(33d)

Having learnt the response of second order systems, we can appreaciate the influence of poles on the system response. The poles which are far away from the origin will not have much influence on the transient response. The complex conjugate poles near the origin ogf the s-plane relative to the other poles dominate the transient response and are called dominant poles. If we have a real pole and complex poles, if the ratio of real pole to the real part of the complex pole exceeds 5, then the complex poles are considered as dominant. Even if the system is of higher order, if there are dominant poles the system can be reduced to second order.



Example 2

For a negative unity feed back system with forward transfer function $G(s) = \frac{25}{s(s+7)}$ find the rise time, peak time, 2% settling time, and maximum percentage overshoot.

Given $G(s) = \frac{25}{s(s+7)}$, $H(s) = 1$

Transfer fuunction of the system= $= \frac{Y(s)}{X(s)} = \frac{G(s)}{1+G(s)H(s)} = \frac{\frac{25}{s(s+7)}}{1+\frac{25}{s(s+7)}} = \frac{25}{s^2+7s+25}$

Comparing this with the standard form $\frac{\omega_n{}^2}{s^2+2\zeta\omega_n s+\omega_n{}^2}$

$\omega_n{}^2 = 25$, and $2\zeta\omega_n = 7$

Hence $\omega_n = 5$ rad/s, and $\zeta = \frac{7}{2\omega_n} = 0.7$ (The system is underdamped)

Rise time $t_r = \frac{\pi-\theta}{\omega_d}$ where $\theta = \cos^{-1}\zeta = 45.6^0 = 0.79\, rad$

$\boldsymbol{\omega_d = \omega_n\sqrt{1-\zeta^2} = 3.57\, rad/s}$

$t_r = \frac{\pi-0.79}{3.57} = 0.659\, s$	$t_p = \frac{\pi}{\omega_d} = 0.88\, s$
$t_s \cong \frac{4}{\zeta\omega_n} = 1.14\, s$	% max overshoot $= e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}} x100 = 4.5\%$

Example 3

A second order control system has the following specifications. Find the required values of the natural frequency and damping ratio of the system. What will be its rise time & peak time?

1. Maximum percantage overshoot permitted is 5%.
2. 2% settling time must be less than $4s$

Given % overshoot =5% $(\omega_d = \omega_n\sqrt{1-\zeta^2} = 1.05 rad/s)$

$(\theta = \cos^{-1}\zeta = 46.4^0 = 0.809\, rad)$

$e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}} = 0.05$ »» $-\frac{\pi\zeta}{\sqrt{1-\zeta^2}} = \ln 0.05 = -2.995$ $\pi^2\zeta^2 = 8.974(1-\zeta^2)$ »» $\boldsymbol{\zeta = 0.69}$	Settling time t_s =4 $\frac{4}{\zeta\omega_n} = 4$ »» $\boldsymbol{\omega_n = 1.45\, rad/s}$
Peak time $t_p = \frac{\pi}{\omega_d} = 2.99\, s$	Rise time $t_r = \frac{\pi-\theta}{\omega_d} = \frac{\pi-0.809}{1.05} =$ $2.22\, s$

#Exercise 13

The open loop transfer function of a unity negative feed back system is $G(s) = \frac{K}{s(s+2)}$. A system response to a unit step input is specified as follows.

Peak time $tp = 1.1s$,

Maximum overshoot$= 5\%$.

Determine whether both the specifications can be met simultaneously.

#Excercise 14

A unity negative feedback control system has the plant $G(s) = \frac{K}{s(s+\sqrt{2K})}$. Determine the percentage overshoot and 2% settling time for unit step input. For what range of K, the settling time is less than 1 s.

#Exercise 15

A closed loop control system has a transfer function $T(s) = \frac{500}{(s+10)(s^2+10s+50)}$. Plot its step response using Matlab/Scilab. Then consider the relatively dominant complex poles and plot its step response. Compare the results.

#Excercise 16

The control system for positioning the head of a cd drive has the closed loop transfer function $T(s) = \frac{0.313\,(s+0.8)}{(s+0.25)(s^2+0.3s+1)}$. Plot using Matlab/Scilab, the poles and zeroes and the step response of the system. What are the time response specifications like rise time, peak time, and overshoot. Now look at the dominance of the complex poles. If we consider the dominannt poles only, how the tansient response specifications change?